

This PDF is a selection from a published volume from the
National Bureau of Economic Research

Volume Title: Immigration, Trade and the Labor Market

Volume Author/Editor: John M. Abowd and Richard B. Freeman, editors

Volume Publisher: University of Chicago Press

Volume ISBN: 0-226-00095-8

Volume URL: <http://www.nber.org/books/abow91-1>

Conference Dates: September 11-12, 1987

Publication Date: January 1991

Chapter Title: The Effects of Immigration on the Labor
Market Outcomes of Less-skilled Natives

Chapter Author: Joseph G. Altonji, David Card

Chapter URL: <http://www.nber.org/chapters/c11773>

Chapter pages in book: (201 - 234)

7 The Effects of Immigration on the Labor Market Outcomes of Less-skilled Natives

Joseph G. Altonji and David Card

One of the most controversial aspects of immigration policy is the extent to which the arrival of immigrants helps or harms less-skilled natives. Although economists have developed a variety of theoretical models to analyze this question (see, e.g., Johnson 1980a, 1980b; Chiswick 1982; or Borjas 1987), relatively little empirical evidence is available.¹ In this paper, we use variation in the fraction of immigrants across different cities to measure the effects of immigration on the labor market outcomes of less-skilled natives. We assemble information from the 1970 and 1980 Censuses on labor market outcomes of natives in 120 major cities. Information from consecutive Censuses allows us to correlate changes in immigrant fractions with changes in native outcomes within cities—thereby abstracting from differences across cities that might bias a simpler cross-sectional analysis. We also provide a variety of information on the industry distributions of natives and immigrants and analyze the changes in these distributions that have occurred in cities with higher and lower immigrant shares.

In the first section of the paper, we present a simple theoretical model that describes the effects of immigration on the domestic labor market. We assume that the labor market within each city consists of skilled and unskilled workers and that immigration adds workers to both sectors, with relative additions

David Card is professor of economics at Princeton University and a research associate of the National Bureau of Economic Research. Joseph G. Altonji is professor of economics at Northwestern University and a faculty research fellow of the National Bureau of Economic Research.

The authors are grateful to Brain McCall and Sarah Turner for assistance with this research and to the National Bureau of Economic Research, the Center for Urban Affairs and Policy Research, Northwestern University, and the Industrial Relations Section, Princeton University, for research funding. They thank John Abowd, Francine Blau, George Borjas, Gregory DeFreitas, Richard Freeman, Peter Kuhn, and participants in seminars at Columbia University, the University of Minnesota, Princeton University, and the National Bureau of Economic Research for comments on earlier drafts.

depending on the nature of immigrant inflows to the city in question. Our theoretical framework departs from earlier models in two ways. On the one hand, we disaggregate labor along skill lines rather than along the lines of national origin. On the other hand, we allow for demand-side effects associated with increases in the local population and for supply-side effects associated with the possible crowding out of native workers in response to lower wage rates. The model leads to a simple empirical specification in which wage and employment outcomes of less-skilled natives (either in cross section or within cities over time) vary with the share and skill composition of immigrants in the local labor market.

In the second section of the paper, we address the question of whether immigrants and natives within the same city compete in the same labor market. Given the size of immigrant flows during the last two decades, our theoretical analysis implies that large adverse effects on less-skilled natives are unlikely unless increases in immigration lead to proportionately larger increases in the supply of labor to less-skilled jobs. We focus on industry-specific labor markets within cities. We develop a simple index that measures the effect of a given inflow of immigrants on the labor market of natives. We find that a 1 percentage point increase in the share of immigrants in a city generates approximately a 1 percent increase in the supply of labor to industries in which less-skilled natives are employed. The degree of competition between immigrants and less-skilled natives varies somewhat by race and sex group, being highest for black females and lowest for black males. Overall, however, the results suggest that immigrants are not sufficiently concentrated in the industries that employ less-skilled natives to have large effects on the less-skilled native groups.

We go on to investigate whether immigrant inflows have displaced less-skilled natives from certain industries. Here, we compare the industry distributions of less-skilled natives in cities with relatively high and relatively low immigrant densities. We find some evidence that less-skilled natives in high-immigrant cities have moved out of immigrant-intensive industries. We also find that the nationwide trend of falling employment in these industries has been slower in high-immigrant cities, suggesting that the availability of immigrant labor has enabled certain low-wage industries to survive in high-immigrant cities.²

In the third section of the paper, we turn to a regression analysis of the relation between immigrant shares (or the change in immigrant shares) and employment outcomes of natives (or the change in these outcomes) across major cities. The results vary somewhat between the cross-sectional and first-difference analyses. We argue, however, that the first-difference analysis is less likely to be contaminated by city-specific factors that affect immigrant densities and native outcomes. The analysis of changes shows no effect of increased immigration on participation or employment rates of less-skilled natives. It does reveal a systematically negative effect on native wages, al-

though the specific estimates depend on the group and on whether we use an instrumental variables procedure to account for the fact that immigration inflows may depend on local labor market conditions. For the four race/sex groups that we consider, the instrumental variables estimates (which we prefer) imply that an inflow of immigrants equal to 1 percent of the population of a standard metropolitan statistical area (SMSA) reduces average weekly earnings of less-skilled natives by about 1.2 percent.³ The least squares estimates, by comparison, imply a more modest .3 percent reduction.

7.1 Analytical Framework

Our framework for analyzing the effect of immigration on the labor market outcomes of less-skilled natives is to view the inflow of immigrants to each city (or, more precisely, SMSA) as an outward shift in the supply of labor. Since we are specifically interested in the effects of immigration on less-skilled natives, we consider a two-sector labor market consisting of skilled and unskilled labor. Within skill categories, we make no distinction between native and immigrant labor or between earlier and later cohorts of immigrants. We assume that the demands for skilled and unskilled labor in each city are decreasing functions of their respective wage rates and that prices of capital and other inputs are exogenous to the local labor market.

This framework contrasts with the one adopted by Borjas (1987), for example, who treats immigrants and natives as separate factors of production and assumes that locally produced output is sold at an exogenous price. In this case, the conventional elasticities of labor demand are undefined since an increase in the wage rate of one type of labor with other factor prices held constant leads to an increase in marginal cost that drives local firms out of business.⁴ Given that many of the goods produced within a city are nontraded services, however, and that many others enjoy some degree of imperfect substitutability due to transportation costs, we believe that it is more reasonable to posit the existence of downward-sloping labor demand functions at the local level.

The observation that the demand for labor within a local economy arises in part from the demand for location-specific goods and services implies that a partial equilibrium model of the labor market is potentially misleading. In the extreme case, if *all* output is locally consumed, and if new immigrants arrive in the same skill proportions as the existing labor force, then an influx of immigrants leads to a new equilibrium at the original wage rates, with proportionately higher levels of employment, output, and consumption.⁵ More generally, the arrival of new immigrants shifts the demand for city output and hence the demand functions for skilled and unskilled labor. The size of this effect depends on the share of output consumed locally and on the relative skill composition of the existing and immigrating labor forces.

To illustrate these propositions and establish a framework for our empirical

analysis, consider an urban economy with two goods: a locally produced good (or service), Y , that is consumed locally and exported to other cities and an imported national good.⁶ Assume that Y is produced by a competitive industry with a constant-returns-to-scale technology using skilled labor, unskilled labor, and other inputs (capital and/or raw materials) whose prices are exogenous and fixed.⁷ Under these conditions, total industry cost (in units of the imported good) is described by a function of the form

$$C(w_s, w_u, Y) = Yc(w_s, w_u),$$

where w_u and w_s represent the real wages of unskilled and skilled labor (in units of the imported good), and $c(\cdot)$ is a unit cost function.⁸ Let q represent the unit price of local output (denoted in units of the imported good). The assumptions of constant returns and perfect competition imply that $q = c(w_s, w_u)$.

Demand for Y arises from three sources: local demand from skilled workers, Y_s ; local demand from unskilled workers, Y_u ; and export demand from the rest of the economy, Y_x . Let $D_s(q, w_s)$ and $D_u(q, w_u)$ represent the per capita demand functions of skilled and unskilled workers, respectively, and let $D_x(q)$ represent the demand function for locally produced output from the rest of the economy. Let P_s and P_u represent the populations of skilled and unskilled workers in the city, and denote the total population by $P = P_s + P_u$. Product market equilibrium requires

$$(1) \quad Y = P_s \cdot D_s(q, w_s) + P_u \cdot D_u(q, w_u) + D_x(q).$$

Let $L_s(w_s, q)$ and $L_u(w_u, q)$ represent the per capita labor supply functions of skilled and unskilled workers, respectively. Equilibrium in the local labor market requires

$$(2a) \quad P_s \cdot L_s(w_s, q) = Y \cdot c_1(w_s, w_u)$$

and

$$(2b) \quad P_u \cdot L_u(w_u, q) = Y \cdot c_2(w_s, w_u),$$

where $c_1(\cdot)$ and $c_2(\cdot)$ denote the partial derivatives of the unit cost function with respect to unskilled and skilled wage rates, respectively.

Suppose that in an initial equilibrium the fraction of unskilled workers in the local population is $a = P_u/P$. We wish to analyze the effect of an inflow of immigrants of size ΔI . Let α represent the share of unskilled workers in the new group. The effects of an immigrant inflow can be obtained by differentiating equations (1), (2a), and (2b) and making use of the fact that the proportional change in the price of output, $\Delta q/q$, equals the share-weighted sum of the proportional changes in all factor prices.

For simplicity, assume that the cross-elasticities of the output demand and labor supply are zero.⁹ Then the proportional changes in skilled and unskilled wage rates satisfy the following pair of equations:

$$(3a) \quad \lambda_u (\alpha/a) \Delta I/P = (\eta_{uu} - \varepsilon_u) \Delta \log w_u + \eta_{us} \Delta \log w_s,$$

$$(3b) \quad \lambda_s [(1-\alpha)/(1-a)] \Delta I/P = \eta_{su} \Delta \log w_u + (\eta_{ss} - \varepsilon_s) \Delta \log w_s,$$

where η_{ij} is the elasticity of labor demand for skill group i with respect to the wage of group j , ε_i is the elasticity of labor supply of group i , and λ_s and λ_u are a pair of numbers between zero and one:

$$\begin{aligned} \lambda_u &= (Y - Y_u - k_1 \cdot Y_s) / Y, & k_1 &= a(1 - \alpha) / [\alpha(1 - a)], \\ \lambda_s &= (Y - k_2 \cdot Y_u - Y_s) / Y, & k_2 &= \alpha(1 - a) / [a(1 - \alpha)]. \end{aligned}$$

The labor demand elasticities in equations (3a) and (3b) are determined by the conventional Marshall-Hicks formulas:

$$\eta_{ij} = \theta_j (\sigma_{ij} - \gamma),$$

where θ_i is the share of the value of output paid as wages to skill group i , σ_{ij} is the partial elasticity of substitution of skill group i with respect to group j , and γ is the elasticity of demand for Y with respect to its relative price q (a weighted average of the elasticities of demand exhibited by consumers in the local market and those elsewhere in the economy).

The expressions $\lambda_u (\alpha/a) \Delta I/P$ and $\lambda_s [(1 - \alpha)/(1 - a)] \Delta I/P$ in equations (3a) and (3b) give the effective percentage increases in unskilled and skilled labor resulting from an inflow of immigrants ΔI . The increases in skilled and unskilled populations are $\alpha \Delta I$ and $(1 - \alpha) \Delta I$, respectively. The proportional increases in the populations of unskilled and skilled workers are therefore $(\alpha/a) \Delta I/P$ and $[(1 - \alpha)/(1 - a)] \Delta I/P$, respectively. The factors λ_u and λ_s adjust the gross increases in labor supply for the net increases in demand generated by the new immigrants. If local output is consumed entirely within the city and immigration is balanced in the sense that $\alpha = a$, then $\lambda_u = \lambda_s = 0$. Otherwise, the effective increases in labor supply depend on the fraction of local output sold outside the city and on the imbalance of skill ratios between the existing and the newly arriving population. In the simple case where newly arriving immigrants have the same skills as the existing population, $\lambda_u = \lambda_s = Y_x/Y$, the fraction of output exported. If newly arriving immigrants are less skilled, however, $\lambda_u > Y_x/Y > \lambda_s$, accentuating the effective increase in unskilled labor supply.

Using equations (3a) and (3b), changes in wages rates can be related to changes in the fraction of immigrants in the local population (f) by noting that $\Delta f = \Delta(I/P) = (1 - f) \Delta I/P$. In the special case that the demand for unskilled labor is independent of the wage rate of skilled labor (i.e., $\eta_{us} = 0$), equation (3a) can be simplified to

$$\begin{aligned} (4) \quad \Delta \log w_u &= \frac{-\lambda_u}{\varepsilon_u - \eta_{uu}} (\alpha/a) \Delta I/P, \\ &= \frac{-\lambda_u}{(1 - f)(\varepsilon_u - \eta_{uu})} (\alpha/a) \Delta f, \end{aligned}$$

which specializes to the formula derived by Johnson (1980a) when $\lambda_u = 1$ and $\alpha = a$.¹⁰ Our model extends Johnson's earlier analysis in two directions: by allowing for skilled and unskilled workers in the existing and immigrating populations and by accounting in a very simple manner for the effect of added population on the demand for local output.

If the demand for unskilled workers depends on the wage rate of skilled labor (i.e., $\eta_{us} \neq 0$), then the expression for the change in unskilled wage rates takes the more general form

$$(5) \quad \Delta \log w_u = B_u \Delta I/P,$$

where

$$B_u = \frac{-\lambda_u (\alpha/a) - \lambda_s \frac{(1 - \alpha)}{(1 - a)} \eta_{us} / (\epsilon_s - \eta_{ss})}{(\epsilon_u - \eta_{uu}) - \eta_{us} \eta_{su} / (\epsilon_s - \eta_{ss})}.$$

Using the labor supply function, the change in the per capita labor supply of unskilled natives can then be written as

$$(6) \quad \Delta \log L_u = \epsilon_u \cdot B_u \Delta I/P.$$

To get some idea of the magnitude of the coefficient B_u relating wage changes to immigrant inflows, suppose that $\alpha = a$, so that $\lambda_u = \lambda_s$. In this case, equation (5) can be rewritten as

$$\Delta \log w_u = \lambda b_u \Delta I/P,$$

where the coefficient b_u ($b_u < 0$) is a function only of the supply and demand elasticities for skilled and unskilled labor, and λ equals the fraction of local production exported to other cities. Values of the coefficient b_u corresponding to alternative values of the supply and demand parameters of the model are displayed in table 7.1. The rows of the table present alternative choices for the ratio between the partial elasticity of unskilled labor with respect to nonlabor inputs (σ_{uk}) and the partial elasticity of skilled labor with respect to nonlabor inputs (σ_{sk}). The share-weighted average of these two elasticities is constrained to equal .6.¹¹ The columns of the table present alternative choices for the partial elasticity of substitution between skilled and unskilled labor (σ_{su}). For each choice of the technological parameters, two values of b_u are reported, corresponding to alternative choices for the elasticities of labor supply: .1 and 1.0. Other parameters in the model are set as follows: the share of skilled labor (θ_s) = .4, the share of unskilled labor (θ_u) = .3, and the elasticity of demand for city output (γ) = -2.5.

The first row of the table presents calculated values of b_u under the assumption that capital is a substitute for unskilled labor and a complement for skilled labor.¹² As Hamermesh (1986, 460-62) has noted in his review of the literature on labor demand, many empirical studies based on the distinction be-

Table 7.1

Predicted Effect of an Increase in Immigration on Unskilled Wage Rates

Ratio of Partial Elasticities of Substitution with Capital (σ_{ks}/σ_{ku}) ^a	Labor Supply Elasticity (ϵ) ^b	Partial Elasticity of Substitution of Skilled for Unskilled Labor (σ_{su})		
		.25	1.0	3.0
1. − .25	.1	. . .	− .31	− .42
	1.0	. . .	− .27	− .30
2. 0	.1	− .27	− .39	− .45
	1.0	− .29	− .30	− .31
3. .5	.1	− .42	− .46	− .48
	1.0	− .32	− .33	− .33
4. 1.0	.1	− .49	− .49	− .49
	1.0	− .34	− .34	− .34

Note: For notation and assumptions, see the text.

^a Share-weighted average of substitution elasticities of skilled and unskilled labor with capital is constrained to equal .6.

^b Labor supply elasticities of skilled and unskilled workers are constrained to be equal.

tween blue-collar and white-collar workers in manufacturing have confirmed this hypothesis. In contrast, the last row of the table presents values of b_u under the assumption that skilled and unskilled labor are equally substitutable with capital.¹³ Despite the wide variation in demand and supply parameters represented in the table, the range of the coefficient b_u is relatively modest: from − .49 to − .27.¹⁴ Under the assumption that immigrants add nothing to the demand for locally produced output (i.e., $\lambda = 1$), these coefficients imply that a 1 percent increase in the population of a city due to an influx of immigrants with the same skill composition as the existing labor force reduces unskilled wages by .3–.5 percent. The implied reduction in the per capita labor supply of natives (and existing immigrants) is proportional to this reduction in wages, multiplied by the elasticity of labor supply. If the elasticity of labor supply is in the range of zero to one, the implied reduction in per capita labor supply of natives is 0–.5 percent.

The magnitude of these predicted effects is dampened by any expansionary effect that immigrants have on the demand for locally produced goods. For example, if one-third of output is consumed locally, then the implied wage effects of a given immigrant inflow are reduced by approximately one-third.¹⁵ Any imbalance in the skill distribution of arriving immigrants, on the other hand, accentuates their effect on the local labor market. In the most extreme case, if newly arriving immigrants are all unskilled and the proportion of skilled workers in the existing labor force is .5, then the predicted value of b_u ranges from − 2.0 to − 1.0, implying roughly two to three times larger effects on unskilled wage rates.

Our empirical strategy in section 7.3 below is to correlate variation in the

share of immigrants in the local labor market with variation in the employment and wage outcomes of less-skilled natives. We interpret the coefficient relating wages to immigrant shares as an estimate of the expression B_u in equation (5) and the coefficient relating employment rates (or participation rates) to immigrant shares as an estimate of the product of B_u and the elasticity of labor supply of unskilled native workers. As the previous discussion makes clear, the value of B_u depends on the nature of immigrant flows to each city and on the characteristics of the demand for output produced in each city. Even ignoring these issues (as we do), it is important to keep in mind the potential endogeneity of immigrant inflows to different cities. If the supply of immigrants is wage elastic, then the covariation across cities between the labor market outcomes of natives and the share of immigrants in the labor market will be a positively biased estimate of the expression B_u . In our analysis, we address this issue with an instrumental variables scheme that isolates the component of immigrant inflows associated with the predetermined characteristics of each city.

Before turning to the empirical work, two limitations of the model deserve discussion. First, the model assumes that the existing native population is immobile. However, one might loosely interpret the supply elasticity of natives to reflect both labor supply changes of the current population of the city and out-migration (or in-migration) of natives to (or from) other cities.¹⁶ If one interprets the intercity mobility of natives as raising the long-run elasticity of labor supply, then one would conclude that migration by natives in response to immigrant inflows would lower the effect of immigration on wages. It would also lower the effect on per capita labor supply of natives, as measured by a variable such as the employment/population ratio.¹⁷ However, intercity migration would imply spillover effects on wages and employment/population ratios in other cities, which we ignore in our empirical work.

Second, the model assumes that the local labor market clears. Within the model, unemployment can be viewed as depending on the wage rate relative to the benefits of being unemployed. This view is most sensible in the long run. Barriers to wage adjustment (such as binding minimum wage levels or fixed welfare benefits) might be expected to strengthen the effect of an increase in immigrants on the employment and unemployment outcomes of natives while weakening the effects on wage levels relative to those implied by equations (6) and (7). The employment effects for natives could be especially large if employers of immigrants are less likely to comply with minimum wage laws or to be unionized.¹⁸

7.2 Industry Distributions of Natives and Immigrants

Our empirical analysis is based on the labor market outcomes of less-skilled natives in 120 major SMSAs in the 1970 and 1980 Censuses. We consider four groups of "less-skilled" natives: white males with less than twelve years

of completed education; white females with less than thirteen years of completed education; black males with less than thirteen years of completed education; and black females with less than thirteen years of completed education. Our data base consists of samples of each race/sex group drawn from the 1/100 Public Use Sample of the 1970 Census and the 5/100 "A" sample of the 1980 Census. A description of our sampling procedures and information on our procedures for matching SMSA definitions between the 1970 and the 1980 Censuses are provided in Appendices A and B.

Table 7.2 provides an overview of our samples of less-skilled natives. The samples are restricted to individuals between the ages of nineteen and sixty-four who report themselves as not in school during the Census week.¹⁹ Because of the age and education requirements, the average age of our less-skilled native groups is close to 40. The average years of complete schooling is less than eight for white male high school dropouts and between ten and eleven for the other groups.

The labor market outcomes that we consider are the labor force participation rate during the Census week; the employment rate during the Census week (measured for those in the labor force in the Census week); the

Table 7.2 Descriptive Statistics for Native Samples

Demographic and Economic Characteristics:	White Male Dropouts		White Females No College		Black Males No College		Black Females No College	
	1970	1980	1970	1980	1970	1980	1970	1980
1. Age	44.3	43.5	40.9	40.8	39.1	37.4	38.7	38.3
2. Education	8.5	8.8	10.6	11.0	9.2	10.2	9.6	10.4
3. Labor force participation rate ($\times 100$)	88.8	81.0	47.3	56.5	83.6	78.4	55.1	59.1
4. Employment rate ($\times 100$)	96.0	91.1	95.6	94.0	94.4	86.9	92.6	87.9
5. Employment population rate Census week ($\times 100$)	85.2	73.7	45.2	53.3	78.9	68.3	51.1	52.1
6. Employment population rate last year ($\times 100$)	91.6	82.9	54.5	61.1	86.7	78.0	60.8	60.1
7. Logarithm of weeks worked last year	3.81	3.75	3.57	3.60	3.77	3.69	3.58	3.60
8. Logarithm of weekly earnings last year (current \$)	4.95	5.52	4.26	4.96	4.61	5.29	4.03	4.90
9. Sample size	84,068	24,925	99,488	81,151	27,779	29,723	34,013	34,540

Note: Samples consist of individuals age 16–64 in 120 major SMSAs. Individuals enrolled in school in Census week are excluded. White male dropouts sample includes individuals with less than 12 years of completed education. Samples for other groups include individuals with less than 13 years of completed education. For further information, see App. A.

employment-population ratio in the Census week; the fraction of people who reported working at any time in the previous year (for simplicity, we refer to this as the employment-population ratio last year); and the logarithms of weeks worked and average weekly earnings during the previous year (measured for those individuals who report positive weeks of work and positive earnings in the previous year). Precise definitions of these outcomes are presented in Appendix A.

The model of the previous section treats the market for less-skilled workers within each city as homogeneous. Even within a particular city, however, the market for less-skilled workers may be segmented along industry lines. If immigrants and natives tend to work in different industries, then the first-round effects of new immigration will be mainly concentrated among existing immigrants. If immigrants tend to work in the same industries as a particular subgroup of natives, however, then the effects of immigration on this subgroup of less-skilled natives will be magnified.

Some simple evidence on the correspondence between industry distributions of natives and immigrants is presented in table 7.3. For the ten two-digit industries with the highest immigrant employment shares and the ten industries with the lowest immigrant shares, this table shows the fraction of each of the four less-skilled native groups in the industry in 1980.²⁰ High-immigrant-share industries include several low-wage manufacturing industries (apparel, leather, furniture, miscellaneous manufacturing, and textiles) as well as low-wage service industries (private household services, hotels and motels, restaurants and bars, and transportation services) and agriculture. Low-immigrant-share industries include the government sector as well as railroads, communications, and several regionally based industries (tobacco, pipelines, coal mining, and oil and gas extraction). A comparison of the second and third columns of the table shows that industries with high or low immigrant shares in 1980 exhibited the same characteristic in 1970, although the immigrant fractions in many industries increased sharply between 1970 and 1980.²¹ The immigrant share of total employment in all industries in our sample of 120 cities increased from 6.0 percent in 1970 to 9.6 percent in 1980.²²

The data in table 7.3 suggest that immigrants are most directly competitive with native women—particularly black women. In fact, the proportion of black females in the ten highest-immigrant-share industries in 1980 was almost as high as the fraction of immigrants in those industries. By comparison, black males are the least concentrated in high-immigrant-share industries and the most heavily concentrated in low-immigrant-share industries.

One way to evaluate the effect of immigration on a particular native group is to calculate the overlap in the industry distribution of the group with the industry distribution of immigrants. Assuming that interindustry mobility costs are large, the effects of immigration on native wages will be directly proportional to the average increase in labor supply to industries in which natives are employed. To formalize this measure, let S_{Ni} represent the share of the native group in the i th industry, let E_i represent the initial level of total

Table 7.3 Distributions of Natives in High- and Low-Immigrant-Share Industries, 1980

Industry	%	%	% of All	% of Natives in Industry				
	Immigrant 1980	Immigrant 1970	Immigrants in Industry	All	White Males	White Females	Black Males	Black Females
High immigrant share:								
1. Apparel	38.4	21.1	5.1	1.3	.6	2.0	.5	2.3
2. Leather	27.3	14.4	.6	.2	.3	.3	.1	.3
3. Agriculture, crops	25.8	10.0	1.5	.6	1.2	.4	.5	.3
4. Furniture	21.0	11.0	1.0	.4	.7	.4	.6	.4
5. Miscellaneous manufacturing	20.9	10.6	2.3	1.1	1.2	1.3	1.0	1.4
6. Private household services	20.2	9.5	1.4	.7	.2	.8	.2	6.0
7. Hotels and motels	18.2	10.6	2.2	1.2	.7	1.7	1.2	3.5
8. Transportation ser- vices	15.8	11.2	.5	.3	.1	.4	.1	.1
9. Restaurants and bars	15.6	9.3	6.4	3.9	2.5	7.6	3.1	5.5
10. Textile mills	15.6	8.8	.8	.5	.7	.7	.6	.8
Total: 10 industries	. . .	. . .	21.8	10.1	8.2	15.6	7.9	20.6
Low immigrant share:								
1. Pipelines	1.5	1.9	.0	.0	.0	.0	.0	.0
2. Gov't.: justice and public safety	2.8	2.3	.4	1.4	.9	.8	2.0	1.0
3. Gov't.: revenue and taxation	2.8	3.4	.1	.4	.0	.5	.2	.5
4. Coal mining	3.5	2.4	.0	.1	.1	.0	.1	.0
5. Railroads	3.8	3.5	.3	.6	1.4	.1	1.1	.2
6. Tobacco	3.9	1.8	.0	.1	.1	.1	.2	.2
7. U.S. Post Office	4.1	2.4	.4	1.0	.9	.4	2.6	1.3
8. Oil and gas extrac- tion	4.2	2.0	.2	.4	.4	.2	.2	.1
9. Communications	4.4	3.1	.8	1.7	.5	2.1	1.1	1.9
10. Gov't.: economic programs	4.5	2.9	.3	.6	.2	.5	.9	.8
Total: 10 industries	. . .	. . .	2.5	6.1	4.5	4.7	8.4	6.0

Note: Based on the industry distributions of 19- to 64-year-olds in 120 major SMSAs in the 1980 Census.
^a All natives include all education groups. Other groups are defined in the note to table 7.2.

employment in industry i , and let ΔE_i represent the increase in labor supply to the i th industry associated with the arrival of a fixed number of new immigrants ΔE . The average proportional increase in labor supply experienced by the native group is

$$\sum_i S_{Ni} \frac{\Delta E_i}{E_i}$$

Suppose that new immigrants sort themselves into industries in the same proportions as existing immigrants. Then $\Delta E_i = S_{Ii} \Delta E$, where S_{Ii} is the share of existing immigrants employed in industry i . Finally, $E_i = S_i E$, where S_i is the share of all workers in industry i , and E is level of total employment in the labor market. Thus, the average proportional increase in labor supply experienced by the native group is $\beta \Delta E/E$, where

$$\beta = \sum_i \frac{S_{Ni} S_{Ii}}{S_i}$$

This expression reduces to one in the case of a homogeneous labor market, in which $S_{Ni} = S_{Ii} = S_i$. In a heterogeneous labor market, however, the average proportional increase in labor supply experienced by a particular native group may be more or less than $\Delta E/E$, depending on the degree of similarity between the industry distributions of immigrants and the native group.

Estimates of this index of labor market competition are presented in table 7.4 for the four groups of less-skilled natives. We have calculated the index separately using the 1970 and 1980 industry distributions of natives and immigrants. We have also calculated the index separately over two subsets of cities: the twenty cities with the highest fraction of less-skilled immigrants in 1980 and the forty cities with the lowest fraction of less-skilled immigrants in 1980. These cities are identified in Appendix D.

Estimates of the index of labor market competition are very similar using the 1970 and 1980 industry distributions. The values of the index range from a low of .85 in 1980 for white males in low-immigrant cities to 1.28 in 1970 for black females and are consistently below one for black males. The results confirm the impression that black females are in most direct competition with immigrants, whereas black males are most isolated from immigrant competition. Nevertheless, the values of the index are not far from one for any of the groups, suggesting that increases in the share of immigrants in the labor market have roughly proportional effects on the labor markets of unskilled natives.²³ The differences in the index between high- and low-immigrant cities are positive for males and negative for females, suggesting that immigrants and native males are in more direct contact in high-immigrant cities while immigrants and native females are in less direct contact. One interpretation of this finding is that less-skilled native females have been displaced from immigrant-intensive industries in high-immigrant cities. We explore this hypothesis next.

Evidence on the extent of industry displacement is presented in tables 7.5 and 7.6, which give the cross-sectional and time-series patterns of differences in the industry distributions of less-skilled natives in high-immigrant and low-immigrant cities. For ten high-immigrant-share industries and ten major immigrant-employing industries, table 7.5 displays the relative share of unskilled natives in high- versus low-immigrant cities. Specifically, let $E_{N_i}^H$ and $E_{N_i}^L$ represent the employment of native group N in industry i in high-immigrant and low-immigrant cities, respectively. Let E_i^H and E_i^L represent

Table 7.4

Estimated Index of Labor Market Competition between Immigrants and Natives

Native Group	All Cities		High-Immigrant Cities		Low-Immigrant Cities	
	1970	1980	1970	1980	1970	1980
1. White male dropouts	1.06	1.00	1.09	1.03	.99	.85
2. White female no college	1.09	1.08	1.05	1.03	1.10	1.12
3. Black males no college	.94	.94	.97	.93	.91	.91
4. Black females no college	1.24	1.15	1.28	1.06	1.20	1.16

Note: For definition of index, see the text. High-immigrant cities include 20 SMSAs with highest fraction of less-skilled immigrants. Low-immigrant cities include 40 SMSAs with lowest fraction of less-skilled immigrants.

total employment in industry i in these cities, and let E_N^H and E_N^L represent total employment of the native group in these cities. For each industry and native group, table 7.5 displays the ratio

$$\frac{E_{N_i}^H / E_i^H}{E_{N_i}^L / E_i^L} \div \frac{E_N^H / E^H}{E_N^L / E^L},$$

which represents the relative employment share of natives in the i th industry in high- versus low-immigrant cities, divided by the relative shares of natives in total employment in those cities. A value of unity indicates that natives have equal shares of employment in the industry in the two groups of cities, controlling for their relative shares in total employment. A value of less than unity, on the other hand, indicates relative displacement in the high-immigrant-fraction cities.

For most of the high-immigrant-share industries, there is evidence of displacement of natives in the high-immigrant-share cities. The displacement effects are less apparent for white males, with ratios in excess of unity for four industries.²⁴ For the other three groups, however, relative employment shares in the set of high-immigrant cities are generally less than unity. By comparison, the evidence of displacement of less-skilled natives from the major immigrant-employing industries in the lower panel of table 7.5 is mixed. On balance, these data suggest that the industry displacement of natives is restricted to low-wage service and manufacturing industries and agriculture. As the ratios in the right-hand column of table 7.5 suggest, these industries are generally more important in high-immigrant than low-immigrant cities, although in cross section it is difficult to distinguish alternative explanations for this effect.²⁵

Table 7.6 repeats the analysis in table 7.5, taking the ratio of the relative

Table 7.5 Relative Industry Distributions of Natives in High- and Low-Immigrant Cities, 1980

Industry	% of All Immigrants in Industry	Relative Share of Native Group: High- vs. Low-Immigrant Cities ^a				High- vs. Low- Immigrant Cities ^b
		White Males	White Females	Black Males	Black Females	
High immigrant share:						
1. Apparel	5.1	1.43	.49	1.29	.44	2.64
2. Leather	.6	1.33	.71	.62	.97	1.40
3. Agriculture, crops	1.5	.56	.86	.84	.74	1.71
4. Furniture	1.0	.64	.68	.68	.36	.94
5. Miscellaneous manufacturing	2.3	.83	1.04	.65	.66	1.89
6. Private household services	1.4	...	.65	.35	.79	1.25
7. Hotels and motels	2.2	1.42	.91	.67	.54	1.25
8. Transportation services	.5	.59	1.12	.09	1.33	2.29
9. Restaurants and bars	6.4	1.32	.80	.95	.50	1.01
10. Textile mills	.8	.73	.77	1.22	.65	.57
Other major immigrant employers:						
1. Hospitals and health services	8.4	1.71	.89	1.48	1.07	.91
2. Construction	5.7	.97	1.04	.83	.81	1.00
3. Education	4.5	.94	1.15	1.07	1.00	.89
4. Business services	3.3	1.51	.81	1.18	.99	1.51
5. Electrical equip- ment	3.3	.75	1.13	.61	.82	1.17
6. Machinery	3.2	.91	1.62	.84	1.32	.68
7. Transportation equipment	2.7	.78	1.52	.74	.72	.74
8. Grocery stores	2.6	1.61	.89	1.89	.98	1.03
9. Wholesale trade: nondurables	2.5	1.27	.94	.96	1.33	1.17
10. Food products	2.1	.81	1.35	.65	.70	.79

Note: Based on the industry distributions of 19- to 64-year-olds in 120 SMSAs in the 1980 Census. High-immigrant cities include 20 SMSAs with the highest fraction of less-skilled immigrants. Low-immigrant cities include 40 SMSAs with the lowest fraction of less-skilled immigrants.

^a For each industry and native group, the relative share is the proportion of industry employment contributed by the native group in high-immigrant cities, divided by the same proportion in low-immigrant cities. This ratio is then divided by the ratio of the shares of the native group in total employment in the two groups of cities.

^b Ratio of industry share of total employment in high-immigrant cities to industry share of total employment in low-immigrant cities.

Table 7.6
Relative Growth of Employment Shares of Natives in High- and Low-Immigrant Cities, 1970–80

Industry	Relative Growth of Native Group: High- vs. Low-Immigrant Cities ^a				Relative Growth of Total Employment: High- vs. Low- Immigrant Cities ^b	Growth of Total Employment All Cities ^c
	White Males	White Females	Black Males	Black Females		
High immigrant share:						
1. Apparel	1.73	.85	.82	.39	1.30	.67
2. Leather	1.33	1.72	.19	.43	3.10	.62
3. Agriculture, crops	.43	.72	1.29	1.45	1.88	.95
4. Furniture	.77	.88	1.26	1.59	1.06	.85
5. Miscellaneous manufacturing	.67	.91	.75	.33	1.11	.96
6. Private household services	. . .	.72	.38	.83	1.55	.52
7. Hotels and motels	1.47	1.15	.71	.72	.93	1.16
8. Transportation services	.61	2.23	.04	2.16	.68	1.39
9. Restaurants and bars	1.36	.97	.89	.98	.94	1.05
10. Textile mills	.94	.88	2.01	.95	.82	.56
Other major immigrant employers:						
1. Hospitals and health services	1.75	1.04	1.00	1.08	.91	1.17
2. Construction	.89	.77	.77	.72	1.13	1.03
3. Education	1.00	1.15	1.27	1.52	.82	.89
4. Business services	1.28	.89	.56	.67	.97	1.32
5. Electrical equipment	.60	1.08	.66	.64	1.38	.75
6. Machinery	.79	1.05	.65	.52	1.40	.93
7. Transportation equipment	.83	1.54	1.07	.87	1.11	.78
8. Grocery stores	1.33	1.04	1.15	1.14	1.07	.92
9. Wholesale trade: nondurables ^d	. . .	. . .	. . .	. . .	. . .	. . .
10. Food products	.75	1.09	.89	.72	.93	.78

Note: For definitions of high-immigrant and low-immigrant cities, see the note to table 7.5.
^a For formula, see the text.
^b Relative ratio of 1980 to 1970 employment totals for industry in high-immigrant vs. low-immigrant cities.
^c Ratio of 1980 to 1970 employment totals for industry in all cities.
^d Data for wholesale trade nondurables industry not available.

employment share of natives in 1980 to the relative employment share in 1970. A value of unity for this ratio suggests that natives have maintained their relative share of industry employment, controlling for the relative growth of total employment of natives in the two sets of cities. A value of less than unity, on the other hand, suggests that natives have lost relative share in the industry in high-immigrant versus low-immigrant cities.²⁶

The results in table 7.6 are generally consistent with those in table 7.5 and suggest some movement of less-skilled natives out of high-immigrant-share industries in the high-immigrant cities between 1970 and 1980. The fifth column of the table indicates the relative growth of total employment by industry in high- versus low-immigrant-share industries, while the sixth column gives the ratio of total employment in the industry in 1980 in all cities to total employment in all cities in 1980. Although several high-immigrant industries were declining relatively quickly between 1970 and 1980, in most cases the relative decline was slower in high-immigrant cities. This suggests that the availability of immigrant labor may allow certain industries to survive in high-immigrant cities even at the same time as natives continue to exit from these industries.

Our analysis of the industry distributions of immigrants and less-skilled natives suggests three conclusions. First, a 1 percentage point increase in the share of immigrants generates approximately a 1 percent increase in the supply of labor to industries in which less-skilled natives are employed. There is no indication that immigrants and less-skilled natives are concentrated in particular industries in a manner that would greatly accentuate the labor market competition between them or, on the other hand, substantially reduce the degree of labor market competition between them. Second, among the four native groups that we consider, immigrants are most directly competitive with black females and least competitive with black men. Third, differences in industry distributions between high- and low-immigrant cities suggest that natives have been displaced from some low-wage service and manufacturing industries and that these industries have declined less quickly in cities with more immigrants.

7.3 An Analysis of the Effects of Immigration on Less-skilled Natives

In this section, we examine the correlation across cities between the labor market outcomes of less-skilled natives and the fraction of immigrants in the city. We present cross-sectional analyses for 1970 and 1980 as well as a first-differenced analysis of changes between 1970 and 1980. Our basic approach is very simple. We regress SMSA averages of the labor market outcome variables for our four race/sex groups against measures of the immigrant fraction in the SMSA and a variety of controls for the characteristics of each city. Before turning to the results of the analysis, however, we first discuss the construction of SMSA means for the outcome variables. We then briefly dis-

cuss potential econometric problems with the cross-sectional and first-differenced analyses and offer some comments on the interpretation of our estimates.

7.3.1 Construction of SMSA-Level Outcome Measures and Control Variables

The first step in our analysis is to construct SMSA-specific means of the outcome variables that are purged of differences in the observable characteristics of the native population across different cities. Given the limited information collected in the Census, this step amounts to regression adjusting the outcome variables for differences in age and education. Such an adjustment has two potential advantages. First, it should reduce the sampling variation associated with the means of the outcome variables across different cities. Second, it should eliminate any bias arising from correlations between the fraction of immigrants in a city and the age and educational attainment of natives.

For each race/sex group in each of the two Censuses, we regress each of the outcome variables against a full set of SMSA dummies and a flexible function of age and education. Specifically, we include a cubic polynomial in age, a detailed set of dummy variables for different education levels, and a full set of interactions of age and education up to the second order. We then use the estimated SMSA dummies as our regression-adjusted outcome measures.²⁷

The explanatory variables in the second step of our analysis include the fraction of immigrants in each SMSA and three additional control variables: the logarithm of SMSA population and SMSA-specific means of age and education for the particular race/sex group under consideration. Although the outcome variables are adjusted for age and education, we found in preliminary work that the mean of adjusted weekly earnings is correlated across cities with the mean of education, particularly for blacks. We have no explanation for this phenomenon, although it may indicate a correlation across cities between the quality and the quantity of education among blacks or possibly a market externality associated with higher levels of education among the less-skilled black population. In any case, we include SMSA-specific means of age and education for the particular race/sex group in all our SMSA-level regressions. These means are calculated directly from our native extracts.

Our measure of the fraction of immigrants in each SMSA is the fraction of foreign-born residents, taken from published tabulations of the 1970 and 1980 Censuses. From the standpoint of the theoretical model, it would be preferable to use the fraction of immigrants in the local labor force. Since our sample sizes for 1970 are too small to provide reliable estimates of the fraction of immigrants in many of the smaller cities, we have relied instead on the published population data. Provided that changes in the immigrant labor force are proportional to changes in the population of immigrants, the use of fraction of immigrants in the population will not affect our results.

7.3.2 Econometric Issues

We next turn to a brief discussion of our estimating equations. We focus on three issues: possible sources of bias in the estimating equations; the interpretation of differences between cross-sectional and first-differenced estimates of the effects of immigration; and the use of weighted least squares in the estimation.

Our cross-sectional estimating equations have the form

$$(7) \quad \hat{Y}_{Nj} = X_{Nj} b + f_j c + e_{Nj},$$

where $\hat{Y}_{Nj}$ is the adjusted labor market outcome for native group N in city j , X_{Nj} is a vector of control variables for the race/sex group and city (the mean of age and education for the group and the logarithm of SMSA population), f_j is the fraction of immigrants in the city, and e_{Nj} is a residual term. Similarly, our first-differenced estimating equations have the form

$$(8) \quad \Delta \hat{Y}_{Nj} = \Delta X_{Nj} b + \Delta f_j c + \Delta e_{Nj},$$

where ΔZ_j refers to the change in the variable Z in city j between 1970 and 1980.

Depending on the choice of outcome measure Y , these equations have the form of equations (5) or (6) derived from our theoretical model. The interpretation of estimates of the coefficient c obtained from equation (7) or (8), however, depends on the nature of the residual terms in these equations. These residuals can be decomposed into two conceptually distinct components: (1) a market-level SMSA effect due to factors other than immigration (e.g., unmeasured characteristics of natives or demand shocks affecting the local economy) and (2) sampling variation arising from the fact that we observe only a sample of natives in each SMSA. Let Y_{Nj} represent the true population value of the outcome variable for natives in city j . Then we may decompose e_{Nj} as

$$e_{Nj} = a_{Nj} + \hat{Y}_{Nj} - Y_{Nj},$$

where a_{Nj} represents the SMSA effect due to factors other than immigration, and $\hat{Y}_{Nj} - Y_{Nj}$ is the component of e_{Nj} attributable to sampling variability. Only if a_{Nj} is orthogonal to the fraction of immigrants in the city will estimates of the coefficient c from the cross-sectional regression (7) yield unbiased estimates of B_u or $\varepsilon \cdot B_u$, as described by equation (5) or (6). In the first-differenced specification, the corresponding requirement is that *changes* in the unmeasured SMSA effects be uncorrelated with changes in the fraction of immigrants in the city between 1970 and 1980.

Clearly, the main advantage of the first-differenced analysis is that it eliminates any bias introduced by city-specific fixed effects that are correlated with the fraction of immigrants in a city and the labor market outcomes of natives. Transitory effects (associated with transitory fluctuations in the demand for the output of specific cities, e.g.) will still lead to biases in the differenced

analysis if they influence the inflow rate of immigrants. Bartel's (1989) recent analysis suggests that economic conditions have a relatively small effect on the destination city chosen by immigrants. Instead, Bartel's findings suggest that immigrants are mainly attracted to cities with large concentrations of previous immigrants from the same country (see also Greenwood and McDowell 1986). Nevertheless, her research leaves open the possibility that the timing and size of immigrant inflows are affected by economic conditions in particular cities.

We attempt to control for any potential correlation between immigrant inflows and local economic conditions in our first-differenced analysis by an instrumental variables procedure. As suggested by Bartel's (1989) work, we use the fraction of immigrants in a city in 1970 to predict the change in the fraction of immigrants over the following decade.²⁸ Immigrant inflows are strongly correlated with the initial fraction of immigrants in a city, and these variables are reasonably strong predictors of the change in immigrant fraction.

In comparing the cross-sectional and first-difference results, one should also keep in mind that the first-difference analysis is more likely to capture the short-run effects of immigration, in which the capital stock and the industry/skill composition of labor demand have not had time to adjust fully. The effects of immigration on per capita employment rates and wages may weaken over time as natives move to other cities or to labor market sectors that are less affected by immigrant competition. Dynamic issues are not addressed in our formal model, but we suspect that the short-run effects of immigration on employment of less-skilled natives will be larger than the long-run effects. The relative magnitude of the short-run and long-run effects on wages depend on whether there are barriers to wage adjustments in the short run. In fact, we find that the cross-sectional estimates of the effect of immigration on employment outcomes of natives are *larger* than the differenced estimates, whereas the opposite is true of the estimated effects on wages. This leads us to suspect that the differences between the cross-sectional and the differenced results are primarily due to correlations between city-specific effects and immigrant shares that are eliminated in first-differences rather than to a distinction between long-run and short-run effects.

A final econometric issue arises from the relatively small samples of black natives in many cities, particularly in our 1970 sample. We restrict our cross-sectional and differenced analysis of each race/sex group to the set of cities for which we have at least thirty group members in both 1970 and 1980. Consequently, we work with a set of ninety-one cities for black males, a set of ninety-four cities for black females, and a full set of 120 cities for white men and women. We also use weighted least squares methods to estimate our equations, using the square root of the number of observations for the race/sex group in the city as a weight. In our first-differenced specifications, we use as a weight $(N_{70}^{-1} + N_{80}^{-1})^{-1/2}$, where N_{70} and N_{80} are the number of observations for the native subgroup in the SMSA in 1970 and 1980, respectively.²⁹ This

weighting scheme assumes that the residual e_{Nj} arises mainly from sampling variability associated with the estimated outcome measure. Even controlling for the covariates in our models, however, the labor market outcomes of different race/sex groups are correlated across cities, suggesting the presence of omitted city-specific effects. We have not adjusted our standard errors or estimation procedures to take account of such error components.

7.3.3 Empirical Results

To provide an introduction and overview of our results, table 7.7 presents weighted least squares estimates of the effects of immigration on the labor market outcomes of the pooled set of four race/sex groups. The estimated equations include unrestricted intercepts for the four groups as well as group-specific coefficients on the means of age and education. The coefficients on the immigrant share variable and the population variable, however, are restricted to be the same across the four native subgroups.

The cross-sectional results for 1970 show significantly negative effects of an increase in immigrant shares on the labor force participation rates and employment rates of less-skilled natives. The results imply that a 10 percentage point increase in the fraction of immigrants in an SMSA would lead to a reduction in the employment/population ratio of less-skilled natives of roughly 2 percent. The employment rate would also fall by 1 percent, implying an increase in unemployment rates of about 1 percent. Among those who work, average weeks per year would fall by about 2 percent.

Table 7.7 Effects of Immigration on Four Groups of Less-Skilled Natives, Pooled Sample (standard errors in parentheses)

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980-70	1980-70 IV ^a
1. Labor force/ population	-.173 (.066)	-.083 (.049)	.080 (.083)	-.102 (.122)
2. Employment/ population	-.240 (.074)	-.054 (.060)	.404 (.097)	.085 (.144)
3. Employment/labor force	-.109 (.036)	.019 (.040)	.461 (.077)	.231 (.113)
4. Fraction worked last year	-.161 (.063)	-.158 (.050)	.090 (.084)	-.246 (.125)
5. Log weeks worked	-.191 (.078)	-.088 (.061)	.232 (.132)	.142 (.193)
6. Log earnings/week	.467 (.165)	.018 (.112)	-.262 (.228)	-1.205 (.342)

Note: All equations included subgroup-specific intercepts, the total population in the SMSA, and the average education and age of the subgroup in the SMSA (with subgroup-specific coefficients). The sample size is 424.

^a Estimated by instrumental variables. The change in the fraction of immigrants in the SMSA is instrumented with the fraction of immigrants in 1970 and its square.

These negative employment effects contrast sharply with the finding that immigration has a positive effect on weekly wages. The estimated coefficient in row 6 implies that a 10 percentage point increase in the immigrant share would lead to a 4.7 percent increase in weekly earnings. Within the context of our model, these results can be reconciled only if the labor supply elasticity of less-skilled natives is negative.³⁰

The 1980 cross-sectional results for the various employment outcomes also indicate a negative effect of immigration, although the estimated coefficients are smaller in magnitude than those for 1970. In the 1980 data, however, the estimated effect of immigrant densities on the average weekly earnings of natives is essentially zero. This gives further reason for caution in the interpretation of the 1970 results.

Weighted least squares estimates of the first-differenced specification are presented in the third column of table 7.7. In contrast to the cross-sectional results, these estimates suggest a modest *positive* effect of the fraction of immigrants on the employment outcomes of natives. The estimated effect on earnings per week is negative ($-.267$) but not statistically different from zero.

Instrumental-variables estimates of the first-differenced specification are presented in column 4. These estimates give an ambiguous picture of the effect of immigration on the employment outcomes of natives. A marginally significant positive effect on the employment rate in the Census week is counterbalanced by a marginally significant negative effect on the employment-population ratio last year. Nevertheless, the instrumented first-differenced results indicate a significantly negative effect of immigration on wages. The coefficient is -1.2 with a standard error of $.242$. The more negative effect associated with the instrumental variables estimation scheme is consistent with the hypothesis that the least squares estimate is positively biased by endogenous immigration inflows.

On balance, the pooled data suggest that the effect of immigrant densities on the employment and participation rates of natives is small and potentially zero. If the instrumented first-differenced specification is taken at face value, however, the effect on wages is apparently negative. For the most part, these conclusions carry over to the detailed results for the four subgroups, to which we now turn.

Results for Individual Race/Sex Groups

Estimates of the relation between immigrant fractions and the labor market outcomes of black males are presented in table 7.8, which has the same format as table 7.7. As in the pooled analysis, the cross-sectional results for black men suggest a negative correlation between the fraction of immigrants and employment outcomes. In the differenced analysis, however, the relation is much less consistent. Likewise, although the 1970 cross-sectional analysis suggests a positive effect of immigration on black male wages, the 1980 cross-sectional results and the differenced results indicate a negative effect.

Table 7.8 **Effects of Immigration on Black Males with Less than Thirteen Years of Education (standard errors in parentheses)**

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980-70	1980-70 IV ^a
1. Labor force/population	-.145 (.126)	-.136 (.084)	-.040 (.170)	-.273 (.240)
2. Employment/population	-.264 (.156)	-.068 (.115)	.658 (.234)	.285 (.234)
3. Employment/labor force	-.165 (.090)	.046 (.098)	.864 (.210)	.623 (.294)
4. Fraction worked last year	-.183 (.100)	-.214 (.081)	.101 (.168)	-.268 (.168)
5. Log weeks worked	-.154 (.121)	-.051 (.111)	-.447 (.252)	.272 (.351)
6. Log earnings/week	.736 (.346)	-.153 (.248)	-.806 (.494)	-1.910 (.706)

Note: All equations include average age and education in the SMSA as well as total population. The sample size is 91.

^a Estimated by instrumental variables. See the note to table 7.7.

Table 7.9 **Effects of Immigration on White Males with Less than Twelve Years of Education (standard errors in parentheses)**

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980-70	1980-70 IV ^a
1. Labor force/population	-.193 (.075)	-.079 (.083)	.066 (.149)	.036 (.231)
2. Employment/population	-.279 (.101)	-.159 (.112)	.349 (.186)	.109 (.289)
3. Employment/labor force	-.107 (.053)	-.110 (.074)	.343 (.134)	.086 (.211)
4. Fraction worked last year	-.151 (.070)	-.215 (.078)	-.145 (.136)	-.609 (.211)
5. Log weeks worked	-.223 (.074)	-.312 (.106)	-.018 (.211)	-.190 (.328)
6. Log earnings/week	-.264 (.201)	-.178 (.212)	-.356 (.406)	-1.103 (.637)

Note: All equations include average age and education in the SMSA as well as total population. The sample size is 120.

^a Estimated by instrumental variables. See the note to table 7.7.

The results for white male dropouts are presented in table 7.9. These results are very similar to those for black males, although the point estimates of the effects of immigration on wages are somewhat smaller in magnitude. Again, the differenced specifications in particular suggest a negative effect of immigrant densities on native wage rates, while the effects on employment and

participation rates are smaller and vary with the precise measure of employment.

The regression results for black females in table 7.10 are of particular interest, given the evidence in section 7.2 that black women are in closer competition with immigrants than the other three groups. Nevertheless, the estimated coefficients for this group are not much different than those for the other groups. The cross-sectional results suggest a small negative effect of immigrant shares on employment outcomes and a modest positive effect on weekly wages. These conclusions are reversed, however, in the first-differenced analysis, which suggests a generally positive effect on employment rates and a negative effect on wage rates. The differenced results for black females are not particularly sensitive to choice of least squares or instrumental variables estimation, although as in previous tables the strongest negative wage effect is obtained by the instrumental variables procedure.

Table 7.11 presents our results for white females. Again, the cross-sectional results for 1970 indicate a negative relation between immigrant shares and employment outcomes, while the differenced analysis indicates much weaker effects. The cross-sectional and first-differenced specifications fit by least squares suggest a positive effect of immigrant shares on wage rates. When the change in immigrant share is instrumented, however, the estimated wage coefficient is negative and consistent with the results for the other native groups.

A check on the wage effects reported for the different native groups in tables 7.7–7.11 is contained in table 7.12. Here, we estimate the same specifications using the wage outcomes of immigrant workers as the dependent variable. We

Table 7.10 Effects of Immigration on Black Females with Less than Thirteen Year of Education (standard errors in parentheses)

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980–70	1980–70 IV ^a
1. Labor force/population	– .216 (.179)	– .063 (.119)	– .154 (.256)	– .221 (.357)
2. Employment/ population	– .221 (.192)	.003 (.128)	.149 (.269)	.032 (.374)
3. Employment/labor force	– .037 (.105)	.073 (.086)	.457 (.186)	.320 (.259)
4. Fraction worked last year	– .165 (.169)	– .127 (.120)	.054 (.272)	– .219 (.379)
5. Log weeks worked	– .247 (.232)	.143 (.143)	.735 (.387)	.217 (.542)
6. Log earnings/week	1.213 (.402)	.533 (.236)	– .838 (.609)	– 1.369 (.848)

Note: All equations include average age and education in the SMSA as well as total population. The sample size is 94.

^a Estimated by instrumental variables. See the note to table 7.7.

Table 7.11 Effects of Immigration on White Females with Less than Thirteen Years of Education (standard errors in parentheses)

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980-70	1980-70 IV ^a
1. Labor force/population	-.037 (.144)	.058 (.097)	.273 (.137)	-.044 (.207)
2. Employment/population	-.095 (.150)	.027 (.105)	.420 (.154)	-.089 (.240)
3. Employment/labor force	-.132 (.058)	-.045 (.045)	.306 (.125)	-.017 (.190)
4. Fraction worked last year	-.047 (.145)	.005 (.098)	.189 (.146)	-.162 (.222)
5. Log weeks worked	-.094 (.170)	-.118 (.110)	.133 (.270)	.335 (.399)
6. Log earnings/week	.667 (.245)	.397 (.132)	.309 (.430)	-.955 (.663)

Note: All equations include average age and education in the SMSA as well as total population. The sample size is 120.

^a Estimated by instrumental variables. See the note to table 7.7.

Table 7.12 Effects of Immigration on Male Immigrant Wages (standard errors in parentheses)

Outcome Variable	Cross-sectional		First-Differenced	
	1970	1980	1980-70	1980-70 IV ^a
1. Log earnings/week (unadjusted)	-.459 (.357)	-.741 (.181)	-.504 (.381)	-.823 (.512)
2. Log earnings/week (adjusted)	.116 (.302)	-.499 (.167)	-.958 (.354)	-1.492 (.481)

Note: Immigrant group includes males age 16-64 not in school in Census week. All equations include average age and education in the SMSA as well as total population. The sample size is 74.

^a Estimated by instrumental variables. See the note to table 7.7.

use two measures of immigrant wages: the mean of actual log weekly earnings for male immigrants and an adjusted mean that controls for the average levels of age and education of immigrants in each city. The results reveal three findings. First, unadjusted mean earnings of immigrants are more strongly correlated in cross section with the fraction of immigrants than mean earnings that have been adjusted for measured skill attributes. This suggests a negative correlation between the skill level of immigrants and their fraction in the population. Second, as we found for the native groups, the instrumental variables estimate of the first-differenced specification leads to the largest negative estimate of the effect of immigrant densities on wages. Finally, the instrumental variables estimates of the effect of immigrant shares on immigrant wages is very similar to the corresponding estimate for native wages. There is no evi-

dence that immigrants have a stronger negative effect on their own wages than on those of less-skilled natives.

Other Results

We estimated many of our least squares models for the 1970, 1980, and differenced samples with a control for the fraction of blacks in the SMSA population. This addition made little difference to the results.

We also reestimated many of our specifications using the fraction of “less-skilled” immigrants in the SMSA population in place of the overall fraction of immigrants in the SMSA population. We defined the fraction of “less-skilled” immigrants as the product of the fraction of immigrants in the SMSA population and the fraction of male immigrants in the SMSA whose predicted earnings are less than the national median for male immigrants (see App. D). The (unweighted) correlation across 120 cities between the “less-skilled” immigrant fraction and the total immigrant fraction is .94 in 1970 and .95 in 1980. The correlation of changes in the two immigrant measures is .82. Perhaps as a result, least squares results using the fraction of less-skilled immigrants are similar to those reported in tables 7.7–7.11. The regression coefficients typically increase in absolute value, reflecting the fact that the scale of the less-skilled immigrant variable is compressed relative to the other variable. It is worth noting that instrumental variables estimates (using the fraction of immigrants in the SMSA in 1970 and its square as instruments) point to a somewhat larger negative effect of the fraction of less-skilled immigrants on the weekly earnings of natives. The coefficients for black males, white males, black females, and white females are -7.0 , -4.8 , -12.9 , and -12.3 , respectively. These estimates are very imprecise, however, perhaps because the correlation between fraction of immigrants in 1970 and the change in fraction of less-skilled immigrants in the SMSA is only .27.³¹

Finally, we reestimated the 1980 cross-sectional specifications and the first-differenced specifications for each of our labor market outcome variables using the SMSA-specific mean of the corresponding labor market outcome for white males age 31–64 with thirteen or more years of schooling as a control variable. We view this approach, which uses the labor market outcomes of highly skilled workers to control for general labor market conditions within each city, as an alternative to our instrumental variables procedure. It is strictly correct only if, in contrast to the implications of our model, immigration has no effect on more highly educated white males. The results from this alternative procedure are generally similar to our ordinary least squares estimates and suggest smaller negative effects of immigration on less-skilled native wages than the instrumental variables procedure.

7.4 Conclusions

This paper presents a variety of evidence on the effects of immigration on the labor market outcomes of less-skilled natives. Working from a simple

theoretical model of a local labor market, we show that the effects of immigration can be estimated from the correlations between the fraction of immigrants in a city and the employment and wage outcomes of natives. We go on to compute these correlations using city-specific outcomes for individuals in 120 major SMSAs in the 1970 and 1980 Censuses. We also use the relative industry distributions of immigrants and natives to provide a direct assessment of the degree of labor market competition between them.

Our empirical findings indicate a modest degree of competition between immigrants and less-skilled natives. A comparison of industry distributions shows that an increase in the fraction of immigrants in the labor force translates to an approximately equivalent percentage increase in the supply of labor to industries in which less-skilled natives are employed. Based on this calculation, immigrant inflows of the magnitude observed between 1970 and 1980 generated 1–2 percent increases in labor supply to these industries in most cities. A comparison of the industry distributions of less-skilled natives in high- and low-immigrant-share cities between 1970 and 1980 shows some displacement of natives out of low-wage immigrant-intensive industries.

We find little evidence that inflows of immigrants are associated with large or systematic effects on the employment or unemployment rates of less-skilled natives. Our estimates of the effect of immigration on native wage rates are sensitive to the choice of specification and estimation procedure. When we consider first-differences between 1980 and 1970 and use an instrumental variables estimation procedure to control for endogeneity of immigrant inflows, we find that a 1 percentage point increase in the fraction of immigrants in an SMSA reduces less-skilled native wages by roughly 1.2 percent. The least squares estimates imply a wage reduction of .3 percent. We point out a number of reasons to prefer the instrumental variables procedure, but additional research, perhaps with the 1990 Census, will be required before one can draw strong conclusions about the response of wages to immigration.

Appendix A

Sampling Procedures and Variable Definitions

Sampling Procedures

Our 1970 samples are drawn from the 1/100 County Group Public Use Sample based on the 5% version of the 1970 Census questionnaire. The sample universe consists of all individuals age 19–64 currently residing in one of 120 SMSAs. (The samples actually contain 121 SMSAs, but, for comparability with the 1980 Census, Dallas and Fort Worth are considered as one SMSA). As described in the text, our analysis is limited to individuals not

currently enrolled in school and in specific race/sex/education and national origin groups from this universe.

Our 1980 samples are drawn from the 5/100 Public Use "A" Sample of the 1980 Census. The sample universe consists of all individuals age 19–64 currently residing in one of 120 SMSAs (adjusted to 1970 boundaries: see App. B). To limit the size of the samples, stratified random samples of individuals meeting the above requirements were drawn by SMSA. Samples of native-born nonblacks (i.e., race coded as white, American Indian, Asian, or other) were drawn to generate approximately twenty-three hundred observations per SMSA for all age/sex/education levels. The samples were then further restricted to two subsets of observations: females with twelve or fewer years of completed education and males with eleven or fewer years of completed education. Samples of native-born blacks were drawn to generate a maximum of 500 observations per SMSA for black females with twelve or fewer years of completed education and 500 observations per SMSA for black males with twelve or fewer years of completed education. One hundred percent samples of foreign-born individuals were taken for all but five large SMSAs, which were sampled with the following probabilities: Chicago, .400; Los Angeles, .170; Miami, .500; New York, .137; and San Francisco, .550.

Labor Market Outcome Variable Definitions

The following labor market outcome variables are defined for all individuals in the sample universe:

- employed in the previous year ($P35 = 0$ in 1970; $P94 = 1$ in 1980);
- in the labor force in the Census week ($P31 = 1, 2, 4, 5$ in 1970; $P81 = 1, 2, 4, 5$ in 1980);
- employed in the Census reference week ($P31 = 1, 2, 4, 5$ in 1970; $P81 = 1, 2, 4, 5$ in 1980).

For individuals in the labor force in the Census week, a fourth variable is defined to be one if the individual was employed in the Census week and zero otherwise.

For individuals who worked in the previous year and who reported strictly positive values for the number of weeks worked in the previous year ($P36 = 0-5$ in 1970; $P95 > 0$ in 1980) and earnings in the previous year ($P37 = 0-500$ in 1970; $P101 > 0$ in 1980), two additional variables are defined: weeks worked in the previous year and earnings per week in the previous year. For 1980, these variables are constructed directly: weeks worked is measured by variable $P95$; and earnings per week is measured by $P101/P35$. (These calculations make no adjustments for allocated responses or truncation of the reported earnings figure.) For 1970, only interval measures of weeks worked and total annual earnings are available. We assigned midpoints of the intervals to the weeks and earnings figures and then constructed earnings per week as the ratio of the assigned values.

Appendix B

Matching SMSA Definitions between 1970 and 1980

The Public Use Samples of the 1970 Census identify 125 individual SMSAs (see pp. 123–26 of the *Description and Technical Documentation for the Public Use Samples of Basic Records from the 1970 Census*). A total of 120 of these are used in our statistical analysis. Four SMSAs were deleted because of difficulty matching between 1970 and 1980 or because of too small sample sizes: Lorain-Elyria, Ohio; Johnstown, Pennsylvania; San Bernadino–Riverside, California; and Wilkes Barre–Hazleton, Pennsylvania. The Fort Worth SMSA was merged with Dallas (see below).

The Census Bureau publication *Geographic Identification Code Scheme* (1983, 11–17) gives a detailed list of changes in the county-level definitions of SMSAs between 1970 and 1980. In most cases, these changes involve the addition of surrounding counties or parts of these counties to the SMSA. The major exceptions are (1) the combination of Dallas and Fort Worth into a single SMSA; (2) the creation of a separate SMSA consisting of Nassau and Suffolk counties of New York State (formerly part of the New York SMSA); and (3) the reclassification of Bergen County, New Jersey, from the Paterson-Clifton-Passaic SMSA to the New York SMSA.

Our general matching strategy was to redefine 1980 SMSA boundaries to the 1970 boundaries. With only a few exceptions, this involved deleting individuals from the 1980 Census file who resided in counties that were classified as part of the SMSA in 1980 but not in 1970. For example, Montgomery County, New York, was added to the Albany-Schenectady-Troy SMSA in 1973. Individuals in this county were therefore deleted from the 1980 file. County-level information for each household is coded in the variable COGRP (location 6–8 of the household record) of the Public Use “A” Sample of the 1980 Census. County group codes are obtained from the 1980 County Group Equivalence File (1980 Census of Population and Housing, Public Use Micro Data Sample, part 77) and Appendix M of the 1980 Census Public Use Microdata Samples Technical Documentation. In most cases, individual counties are identified by one or more county group codes. For these cases, the deletion is accomplished by specifying the county group code(s) of those counties added to the SMSA after 1970.

In some cases, only parts of a surrounding county group were added to the SMSA. In these cases, we randomly deleted a fraction of individuals from the added county or county group. The fraction of individuals deleted was set equal to the relative population of the part of the county added to the SMSA. Estimates of population for county subgroups were obtained from the 1980 County Group Equivalency File.

In all, a total of forty-nine counties or county subgroups were deleted from the definitions of the 120 SMSAs. Another forty counties or county subgroups

were partially deleted. The number of individual records actually affected by these deletion procedures is small. For example, of 244,941 immigrants identified on the 1980 Public Use A Sample using the 1980 SMSA definitions, 2,609 (1.07 percent) were deleted in the change to the 1970 definitions. A copy of the computer instructions that performed the deletions is available from David Card on request.

To account for changes in the classification of Nassau and Suffolk counties in New York State, we added individuals in the Nassau-Suffolk SMSA in 1980 to the New York SMSA sample. To account for the changes in definition of the Paterson-Clifton-Passaic SMSA, we added individuals in the 1980 sample living in Bergen County, New Jersey (classified as part of the New York SMSA in 1980), to the Paterson-Clifton-Passaic SMSA sample and deleted them from the New York SMSA sample. To account for the reclassification of Dallas and Fort Worth into a single SMSA, we combined individuals from the Dallas and Fort Worth SMSAs in the 1970 Census file into a single Dallas-Fort Worth sample. No attempt was made to deal with minor reclassifications affecting the Boston and Providence SMSAs and the Detroit and Flint SMSAs.

Appendix C

Industry Definitions

Matching of 1970 and 1980 Three-Digit Codes

Our procedure was to reclassify the three-digit industry codes of individuals in the 1970 Census to 1980 industry codes. The Census Bureau provided us with cross-tabulations of 1970 and 1980 three-digit industry codes for samples of males and females who had been coded under both systems. These cross-tabulations were used to estimate the probability that an individual with a given 1970 code would be classified in a particular industry under the 1980 coding scheme. Using these probabilities, a computer program was developed that reclassifies individuals probabilistically from their 1970 three-digit industry to a particular 1980 three-digit industry. The computer program processes males and females separately. A copy of the program is available from David Card.

Industry Classifications Used in Tables 7.3–7.6

Using the three-digit industry titles in Appendix H of the Public Use Microdata Samples Technical Documentation, we developed a “two-digit” classification consisting of seventy-six individual industries. (There are 231 separate industries in the 1980 Census industry coding system.) This classification combines many smaller three-digit industries: for example, “agricultural ser-

vices except horticulture" (industry 020) and "horticultural services" (industry 021). A listing of the computer instructions used to classify three-digit industries into this two-digit system is available from David Card.

Appendix D

Classification of High- and Low-Immigrant Cities

In order to determine average immigrant skill levels by SMSA, a regression equation was fit to the log of average weekly earnings for the 1980 sample of male immigrants. The equation included the same flexible function of age and education used to regression adjust native outcomes (see the text description) as well as a set of forty-six country/region dummy variables and their interactions with an indicator variable for having entered the United States after 1970 and a variable representing years in the United States. (Chiswick [1978], Borjas [1985, 1987], and others have shown that country of origin, immigration cohort, and years since immigration affect earnings in the United States.) This equation was then used to assign a predicted wage to each male immigrant. Immigrants with a predicted wage less than the median predicted wage for the entire United States were classified as "low skill." Finally, the fraction of low-

Table 7D.1 **Twenty Cities with Highest Fraction of Low-Skill Immigrants**

City	Fraction Immigrants	Fraction Low-Skill Immigrants
Miami	.36	.20
El Paso	.21	.20
Los Angeles	.22	.16
Salinas	.19	.16
Jersey City	.24	.15
Oxnard-Ventura	.13	.10
New York	.21	.10
Honolulu	.15	.10
Paterson	.15	.09
Fresno	.11	.09
San Diego	.13	.08
Anaheim	.13	.08
Bakersfield	.09	.08
Stockton	.11	.08
Santa Barbara	.12	.07
San Francisco	.16	.07
San Jose	.14	.07
Houston	.08	.06
San Antonio	.07	.06
Providence	.09	.06

Table 7D.2

Forty Cities with Lowest Fraction of Low-Skill Immigrants

City	Fraction Immigrants	Fraction Low-Skill Immigrants
Huntington-Ashland, KY	.01	.00
Chattanooga	.01	.00
Birmingham	.01	.00
Knoxville	.01	.00
York, PA	.01	.00
Canton	.02	.00
Jackson, MS	.01	.00
Cincinnati	.02	.01
Dayton	.02	.01
Flint	.03	.01
Appleton	.02	.01
Louisville	.01	.01
St. Louis	.02	.01
Nashville	.01	.01
Indianapolis	.02	.01
Richmond	.02	.01
Duluth	.03	.01
Memphis	.01	.01
Akron	.03	.01
Greensboro	.01	.01
South Bend	.03	.01
Utica-Rome, NY	.04	.01
Erie, PA	.03	.01
Pittsburgh	.03	.01
Harrisburg	.02	.01
Binghamton	.04	.01
Greenville	.02	.01
Peoria	.02	.01
Wilmington	.03	.01
Fort Wayne	.02	.01
Mobile	.01	.01
Madison	.03	.01
Lancaster	.02	.01
Toledo	.03	.01
Youngstown	.04	.01
Lansing	.03	.01
Columbus	.02	.01
Atlanta	.02	.01
Minneapolis	.03	.01
Shreveport	.02	.01

skill immigrants in each SMSA was determined by multiplying the fraction of immigrants in the SMSA by the fraction of immigrants who are classified as low skill. Table 7D.1 lists the twenty cities with the highest fraction of low-skill immigrants. Table 7D.2 lists the forty cities with the lowest fraction of low-skill immigrants.

Notes

1. Most of the available evidence is summarized by Greenwood and McDowell (1986), General Accounting Office (1988), and Papademetriou et al. (1989). Two studies of particular relevance to ours are Grossman (1982) and Borjas (1987). Lalonde and Topel (in this volume) provide a parallel study to ours, focusing on the effects of recent immigrants on the labor market outcomes of earlier immigrants. Muller and Espenshade (1985) analyze the effect of immigrants on various California cities.

2. A similar conclusion is reached by Kuhn and Wooton (in this volume) and Papademetriou et al. (1989, ch. 4).

3. The average change in the percentage of immigrants between 1970 and 1980 in the 120 SMSAs in our sample is 1.4 and ranges between 0 and 11.4 percent.

4. If the price of output is exogenous, it is more convenient to work with the elasticities of factor prices with respect to factor quantities, holding constant marginal cost. These are usually known as elasticities of complementarity (see, e.g., Hamermesh 1986).

5. This depends, of course, on constant returns to scale and on perfectly elastic supplies of capital and other inputs.

6. In order to avoid the theoretical prediction of factor price equalization across cities, it is necessary to assume that the number of goods produced within a city is less than the number of locally supplied factors. For further discussion of this point, see Kuhn and Wooton (in this volume).

7. We ignore land or any other locally supplied factors.

8. For notational simplicity, we suppress the dependence of $c(\cdot)$ on the prices of nonlabor inputs.

9. In the notation of eqq. (1) and (2), $\partial D_j(q, w_j)/\partial w_j = 0$, and $\partial L_j(w_j, q)/\partial q = 0$, for $j = (u, s)$.

10. Johnson (1980a) makes the further assumption that the elasticity of labor supply among existing immigrants is zero, so that the effective supply elasticity in the market for unskilled labor is $(1 - f_u)\epsilon$, where f_u is the fraction of immigrants in the existing pool of unskilled workers, and ϵ is the labor supply elasticity of natives.

11. That is, $\theta_u\sigma_{uk} + \theta_s\sigma_{sk} = .6(\theta_u + \theta_s)$, where θ_j represents the value share of labor in the j th skill group.

12. No entries are included in the first row under the column for $\sigma_{su} = .25$. In this row of the table, σ_{sk} is strongly negative ($-.525$). Thus, skilled and unskilled labor must be relatively strong substitutes (i.e., $\sigma_{su} > .8$) to satisfy the restrictions on the matrix of partial elasticities.

13. If $\sigma_{uk} = \sigma_{sk}$, eq. (5) implies that the value of the coefficient b_u is independent of the substitutability between skilled and unskilled labor.

14. The elasticities of demand for unskilled labor with respect to its own wage rate (η_{uu}) implied by the parameter choices in table 7.1 range from -1.0 (in the lower-left-hand entries of the table) to -2.6 (in the upper-right-hand entries of the table).

15. Estimates of the fraction of output produced in a city that is consumed locally are not easily obtained. Roughly 35 percent of consumer expenditures are allocated to personal, health, business, and education services, public utilities, transportation services, and other goods with a high local content.

16. If the immigrants are primarily unskilled, then one might expect out-migration of unskilled natives and in-migration of skilled natives.

17. Filer (1988) shows that the net migration rate of natives to an SMSA between 1975 and 1980 is negatively related to the migration rate of immigrants into the SMSA between 1970 and 1974 and to the migration rate of immigrants into the SMSA be-

tween 1975 and 1980. The negative relation appears to be strongest for low-skilled and less-educated natives.

18. Papademetriou et al. (1989, chap. 4) summarize evidence from a few industry studies suggesting that in some cases immigrant labor has been used to undercut union firms paying higher wages and employing native workers.

19. By "Census week" we mean the week immediately preceding the administration of the Census, for which individuals report their major activity. The Census is administered on 1. April.

20. Our two-digit industry classification is explained in App. C.

21. Of the ten highest-immigrant-share industries in 1980, seven were in the top ten industries by immigrant share in 1970. The rank-order correlation across industries between the 1970 and 1980 immigrant shares is .86.

22. The average fraction of immigrants in the total population in our sample of cities in 1970 was .044 and ranged from .003 to .242. The average fraction of immigrants in the total population in 1980 was .058 and ranged from .008 to .357.

23. It should be pointed out that the index is computed from the industry distribution of existing immigrants and cannot be used to assess the effects of an inflow of immigrants that are much different from the existing stock.

24. The number of white males in private household services is so low that the index cannot be calculated.

25. For example, many high-immigrant-share cities are also major transportation centers (New York, Los Angeles, Miami). This fact may partially explain the relatively high share of the transportation services industry in the high-immigrant-share cities.

26. It is interesting to note that total employment growth rates between 1970 and 1980 for the twenty high-immigrant-share cities and the forty low-immigrant-share cities were virtually identical: the ratio of 1980 to 1970 employment was .92 for the high-immigrant-share cities and .91 for the low-immigrant-share cities. The relative growth rates of less-skilled native employment, however, were somewhat different in the two sets of cities. The relative ratios of 1980 to 1970 employment totals in high-versus low-immigrant cities were .96 for white males, .90 for white females, 1.02 for black males, and .87 for black females.

27. A similar approach is used by Borjas (1987).

28. An alternative strategy is to study the effect of immigrant flows to particular SMSAs that one can identify as exogenous. For example, Card (1990) examines the effect of the Marial boat lift on the Miami labor market and finds little effect on the wages and unemployment rates of less-skilled blacks and other non-Cuban groups. His results for wages are somewhat at variance with the instrumental variables estimates we report below.

29. The instrumental variables estimation of the first-difference equation also uses these weights.

30. The implied per capita labor supply elasticity is roughly minus one. An alternative explanation, which might be consistent with an extended version of the model allowing for heterogeneity within the population of less-skilled natives, is that a downward shift in the wage distribution induced by immigration results in the exit from the labor force of natives with the lowest skill levels. However, given that the decline in the employment population ratio is small, a compositional shift cannot explain the results even if the wages of those who left employment were essentially zero prior to their departure.

31. In contrast, the correlation between the fraction of immigrants in 1970 and the change in fraction of all immigrants in the SMSA is .60. These correlations refer to the unweighted sample of 120 SMSAs.

References

- Bartel, Ann. 1989. Where do the new U.S. immigrants live? *Journal of Labor Economics* 7 (October):371-91.
- Borjas, George. 1985. Assimilation, changes in cohort quality, and earnings of immigrants. *Journal of Labor Economics* 3(October):463-89.
- . 1987. Immigrants, minorities, and labor market competition. *Industrial and Labor Relations Review* 40(April):382-93.
- Card, David. 1990. The impact of the Mariel boatlift on the Miami labor market. *Industrial and Labor Relations Review* 43(January):245-57.
- Chiswick, Barry. 1982. The impact of immigration on the level and distribution of economic well-being. In *The gateway: U.S. immigration issues and policies*, ed. Barry Chiswick. Washington, D.C.: American Enterprise Institute.
- Filer, Randall. 1988. The impact of immigrant arrivals on migratory patterns of native workers. Typescript, Department of Economics, Hunter College-CUNY.
- General Accounting Office. 1988. *Illegal aliens: Influence of illegal workers on wages and working conditions of legal workers*. Washington, D.C.: U.S. Government Printing Office.
- Greenwood, Michael, and John McDowell. 1986. The factor market consequences of U.S. immigration. *Journal of Economic Literature* 24(December):1738-72.
- Grossman, Jean. 1982. The substitutability of natives and immigrants in production. *Review of Economics and Statistics* 64(November):596-603.
- Hamermesh, Daniel. 1986. The demand for labor in the long run. In *Handbook of labor economics*, ed. Orley Ashenfelter and Richard Layard. Amsterdam: North-Holland.
- Johnson, George. 1980a. The labor market effects of immigration. *Industrial and Labor Relations Review* 33(April):331-41.
- . 1980b. The theory of labor market intervention. *Economica* 47(August):309-30.
- Muller, Thomas, and Thomas Espenshade. 1985. *The fourth wave: California's newest immigrants*. Washington, D.C.: Urban Institute Press.
- Papademetriou, Demetrios, et al. 1989. The effects of immigration on the U.S. economy and labor market. U.S. Department of Labor Bureau of International Labor Affairs Immigration Policy and Research Report no. 1, May.